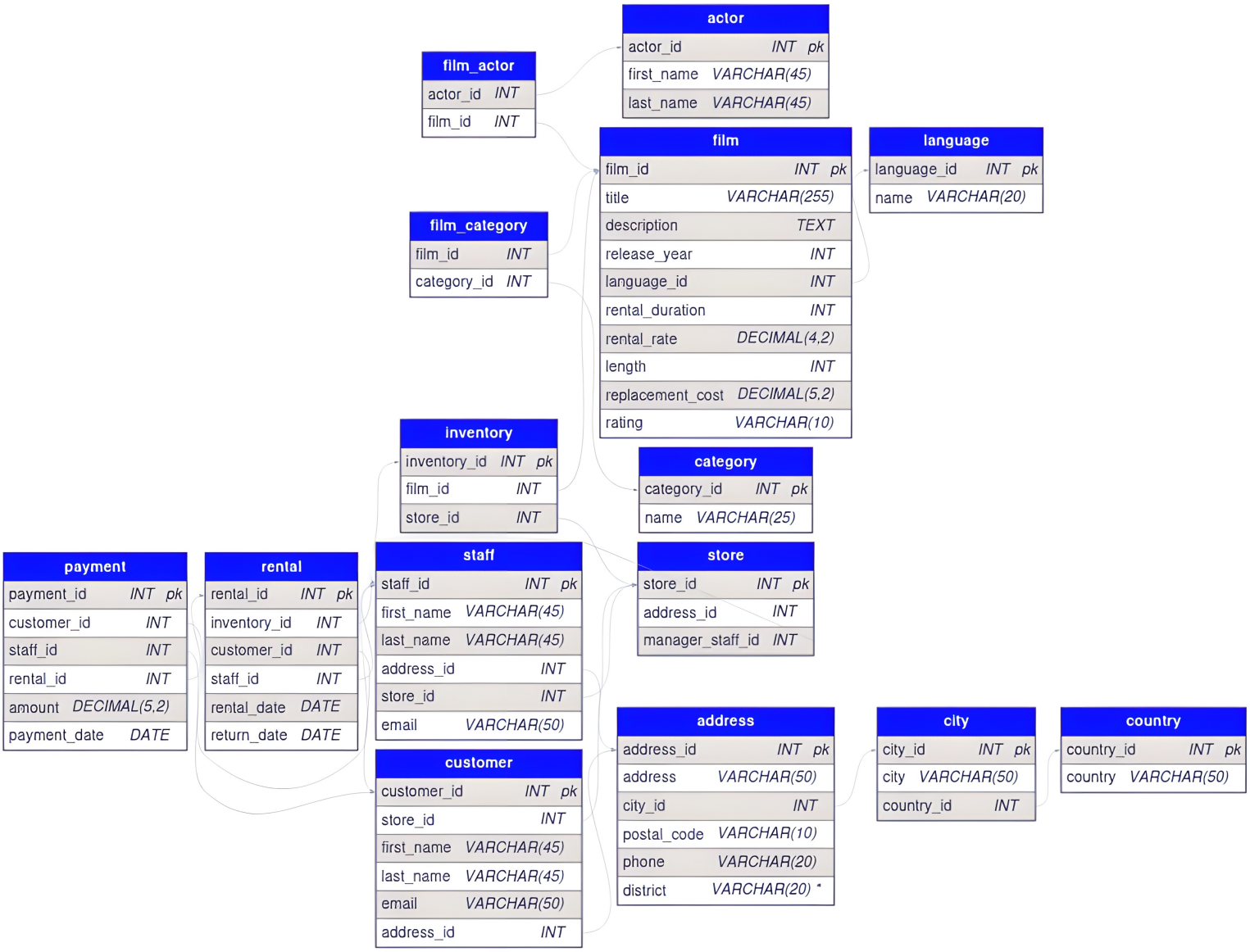
Lab Mid

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

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DBMS Lab Mid



1. Find distinct customers who have made payments of amount 0.

   Solution:

```
SELECT DISTINCT c.customer_id, c.first_name, c.last_name, c.email, p.amount
FROM customer c
JOIN payment p ON c.customer_id = p.customer_id
WHERE p.amount = 0;
```

2. Display only the username part of staff emails (everything before @).

   Solution:

```
SELECT SUBSTR(email, 1, POSITION('@' IN email) - 1) AS username
FROM staff;
```

3. For each customer, count the total number of rentals they have made, sorted in descending order.

   Solution:

```
SELECT c.first_name, c.last_name, COUNT(r.rental_id) AS total_rentals
FROM customer c
LEFT JOIN rental r ON c.customer_id = r.customer_id
GROUP BY c.customer_id
ORDER BY total_rentals DESC;
```

4. List the first name and last name of staff members along with the total amount collected by each in payments.

   Solution:

```
SELECT s.first_name, s.last_name, SUM(p.amount) AS total_collected
FROM staff s
LEFT JOIN payment p ON s.staff_id = p.staff_id
GROUP BY s.staff_id;
```

5. Identify the top 5 customers who spent the highest total payment amount.

   Solution:

```
SELECT c.first_name, c.last_name, SUM(p.amount) AS total_paid
FROM customer c
JOIN payment p ON c.customer_id = p.customer_id
GROUP BY c.customer_id
ORDER BY total_paid DESC
LIMIT 5;
```

6. List rentals that were returned more than 7 days after the rental date. Show rental_id, rental_date, return_date, and total number of days rented.

   Solution:

```
SELECT rental_id, rental_date, return_date, return_date - rental_date AS days_rented
FROM rental
WHERE return_date - rental_date > INTERVAL '7 days';
```

7. For each staff member, show their full name and the latest rental date they handled.

   Solution:

```
SELECT s.first_name || ' ' || s.last_name AS staff_name, MAX(r.rental_date) AS latest_rental
FROM staff s
JOIN rental r ON s.staff_id = r.staff_id
GROUP BY s.staff_id;
```

8. Display the title, rating, and rental rate of all films whose rental rate is greater than 4.00 and whose rating is either “PG” or “PG-13”.

   Solution:

```
SELECT title, rating, rental_rate
FROM film
WHERE rental_rate > 4.00 AND rating IN ('PG', 'PG-13');
```

9. Find all customers who live in a city that belongs to the country `India`, showing the customer’s first and last name, city and country.

   Solution:

```
SELECT
    c.first_name,
    c.last_name,
    c.email,
    ci.city,
    co.country
FROM customer c
JOIN address a ON c.address_id = a.address_id
JOIN city ci ON a.city_id = ci.city_id
JOIN country co ON ci.country_id = co.country_id
WHERE co.country = 'India';
```

10. List the titles of films in the `Comedy` category that have a `length` greater than the average duration of all `Comedy` films.

Solution:

```sql
SELECT f.title, f.length
FROM film f
JOIN film_category fc ON f.film_id = fc.film_id
JOIN category c ON fc.category_id = c.category_id
WHERE c.name = 'Comedy'
AND f.length > (
    -- Subquery: Calculate average length of Comedy films
    SELECT AVG(f2.length)
    FROM film f2
    JOIN film_category fc2 ON f2.film_id = fc2.film_id
    JOIN category c2 ON fc2.category_id = c2.category_id
    WHERE c2.name = 'Comedy'
);
```